

Thermal-power budget of the demonstration node

The demonstration node stores 12000 mJ of usable energy and keeps a reserve of 2000 mJ at every boundary. Load segments are declared as intervals; supply bounds come from the panel datasheet.

Measured decode energy per token was 0.62 mJ on the reference device. The temperature of the radiator plate must stay below 45 °C during downlink.

Table 1 lists the declared segments used by the temporal analysis.

Table 1. Declared segments

segment	duration_s	power_mW
eclipse	1800	450
sunlit	3600	120
downlink	600	980